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# Generating and testing flying focus laser pulses with Lasy for PIconGPU simulations

— A Bachelors Defense —

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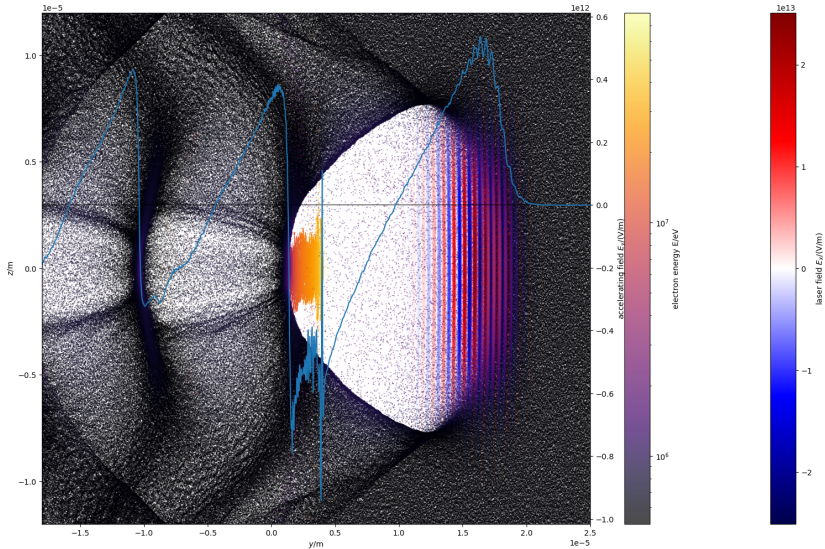
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February 16, 2026



# 1 Dephasingless Laser WakeField Acceleration (DLWFA)

# Laser WakeField Acceleration (LWFA)



Electric field and electrons in an LWFA (see [1]) simulation.

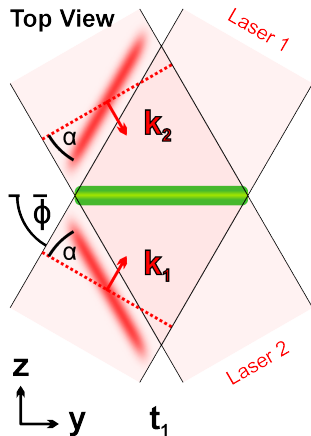
# The problem of dephasing

# Flying focus lasers – solving the problem of dephasing

## 1. TWEAC

- **T**raveling-**W**ave **E**lectron **A**Ccelerator
- Developed by Debus et al [2]
- Uses two laser pulses with tilted pulse fronts
- The tilt controls the velocity of the overlapping region

Image: TWEAC setup using two laser pulses. Image taken from Debus [3]

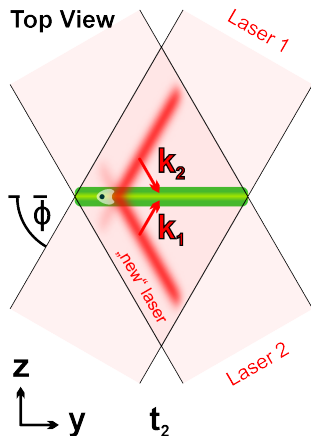


# Flying focus lasers – solving the problem of dephasing

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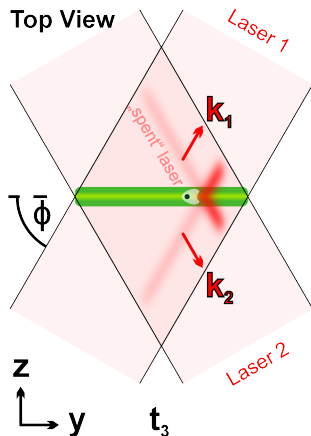


# Flying focus lasers – solving the problem of dephasing

## 1. TWEAC

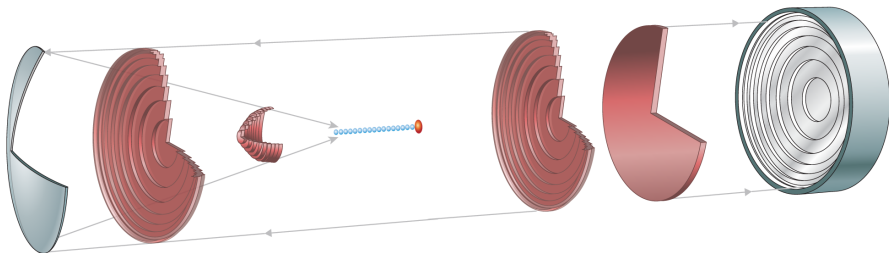
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# Flying focus lasers – solving the Problem of dephasing

## 2. Axiparabola laser



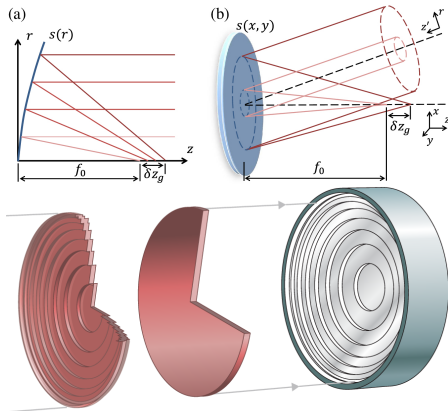
The flying focus setup of Palastro et al [4]. Two optical elements: The Axiparabola (left) and the Radial Group Delay echelon (RGD) (right). Image taken from Palastro et al [4].



# Flying focus lasers – solving the Problem of dephasing

## 2. Axiparabola laser

- Axiparabola [5]:
  - Near-parabolic mirror
  - Focuses light onto a line – the focus region
  - Light at radius  $r$  is focused at  $f(r) = f_0 + \delta \left( \frac{r}{R} \right)^2$
- Radial Group Delay echelon (RGD) [6][4]:
  - Stepped concentric mirror rings
  - Shape follows some function  $\tau_D(r)$
  - controls the timing of the axiparabola focus



Images: Top: Axiparabola functionality. Image taken from Smartsev et al [5]. Bottom: RGD functionality. Image taken from Palastro et al [4].



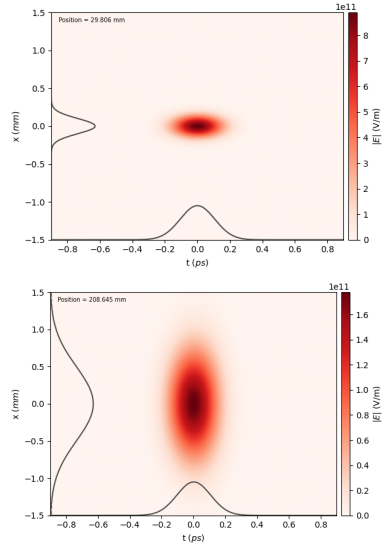
## 2 Flying focus lasers in Lasy and PIConGPU

# Lasy

A python library

- A python library for simulating Laser pulses in a vacuum [7]
- Uses complex envelope of the laser field
- Uses angular spectrum propagation
- Can use cylindrical coordinates for memory and CPU time efficiency
- Offers a range of optical elements

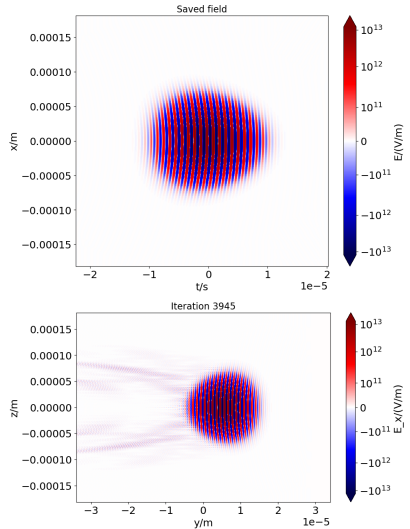
Images: Example of a Gaussian pulse being propagated by Lasy. Top: Generated at the focus. Bottom:  $6 z_R$  after the focus.



# Importing to PIConGPU

- New module `full_field`
- Generates full electric field and saves it using `openPMD-api`
- `incidentField` method  
`FromOpenPMDPulse` (see Dietrich [8]) imports the field into PIConGPU

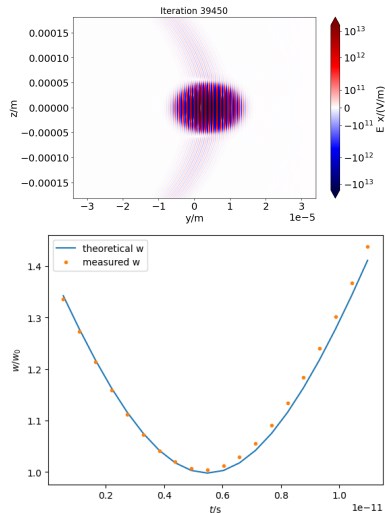
Images: Top: Electric field of a Gaussian laser pulse,  $1z_R$  before the focus, as saved to the file. Bottom: That same electric field entering the simulation window of a PIConGPU simulation.



# Testing the method

- Test with Gaussian laser pulse and parabolic mirror
- Some artifacts are visible
  - Problem of the incidentField method
    - See [9]
- Test successful

Images: Top: Electric field of the Gaussian laser pulse at the focus according to the PIconGPU simulation. Bottom: The beam waist  $w$  of the pulse over time, compared to theory.



# Implementing the flying focus

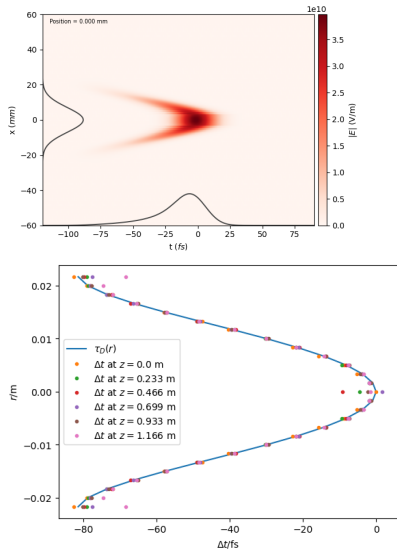
## 1. The radial group delay echelon (RGD)

- Implemented from scratch as Lasy optical element
- Following the description by Ambat et al [6]
- Shapes the pulse temporally without focusing or defocusing
- Can generate any radially symmetric shape of delay  $\tau_D(r)$

Images: A Gaussian pulse after interacting with the RGD.

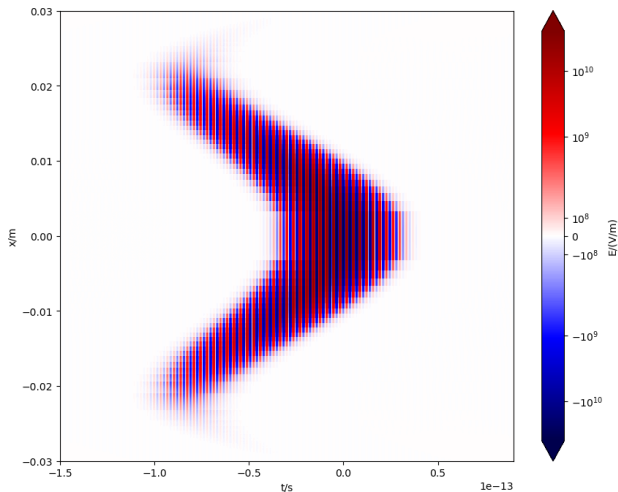
Top: Field envelope. Bottom: Test results. Even after long distances the shape still holds. For comparison:

$z_R \approx 290$  m



# Implementing the flying focus

## 1. The radial group delay echelon (RGD)



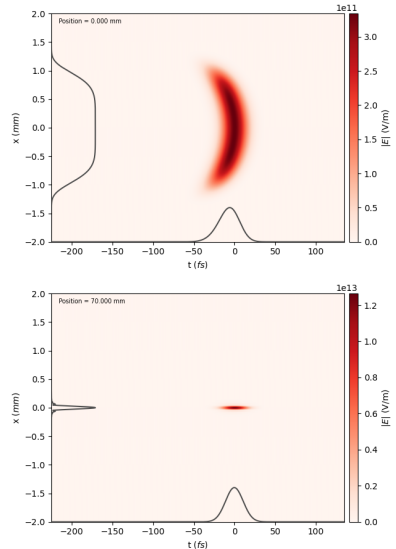
The electric field of the laser after interacting with the RGD.

# Implementing the flying focus

## 2. The axiparabola

- Included in Lasy
- Following Smartsev et al [5]
- Focusing laser pulse in the focus region
- Also implemented an axiparabola following Ambat et al [6]
  - Very small differences

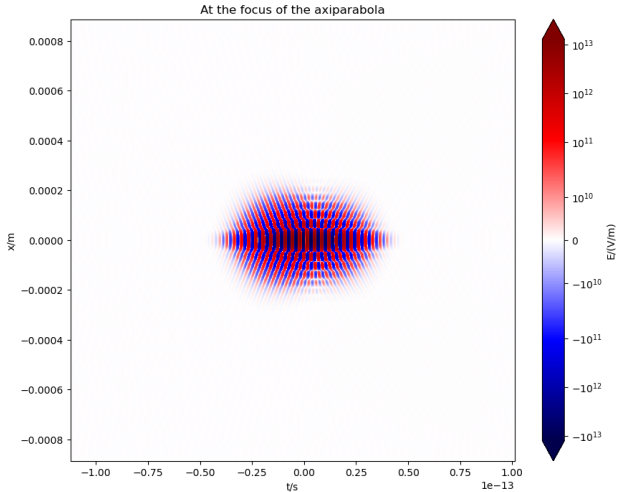
Images: A super-Gaussian laser pulse after reflecting off the axiparabola. Top: In the near field. Bottom: In the far field at the beginning of the focus region.





# Implementing the flying focus

## 2. The axiparabola



The electric field of the laser at the beginning of the focus region of the axiparabola.

# Implementing the flying focus

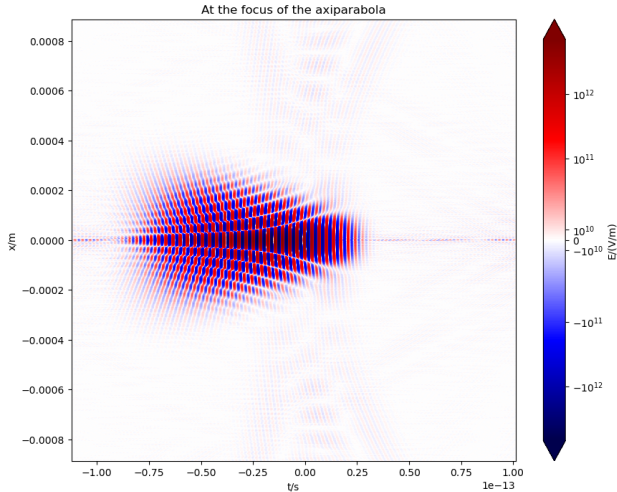
## The complete setup

The simulation had the following steps:

- 1 Initialising the laser pulse with its pulse profile
- 2 Applying the RGD to the laser
- 3 Applying the axiparabola to the laser
- 4 Propagating the laser to the focus region
  - At this point the laser field can be saved to file
  - The next steps can happen in Lasy or PIConGPU
- 5 Measuring the relative focus position  $t_{focus} - f_0/c$  or  $z_{focus} - t \cdot c$  and the beam waist  $w$
- 6 Propagating the laser some distance through the focus region
- 7 Repeating steps 5 and 6 until the end of the focus region is reached

# Implementing the flying focus

## The complete setup



The electric field of the laser at the beginning of the focus region of the complete setup.



### 3 Verifying the flying focus

# Verifying the flying focus

## The measurement

- Relative focus position  $t_{focus} - f_0/c$  (Lasy) or  $z_{focus} - t \cdot c$  (PIConGPU):
  - A central line-out is taken from the field energy, already in relative coordinates
  - Then  $t_{focus} - f_0/c$  or  $z_{focus} - t \cdot c$  is calculated as the expected value of the resulting distribution
- Beam waist  $w$ :
  - The field energy is longitudinally and azimuthally summed
  - Then  $w$  is calculated from the statistical standard deviation of the resulting distribution

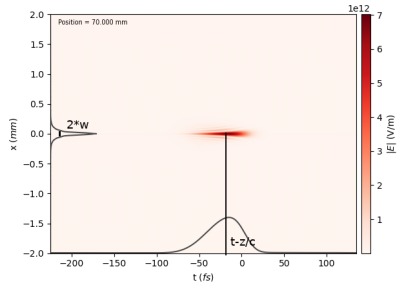


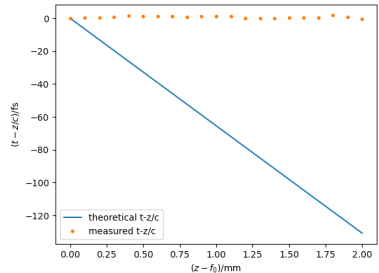
Image: An example of where the measurements can be found in the plots.

# Verifying the flying focus

## First results

- Simulation in Lasy
  - Result: The flying focus effect cannot be seen
- Test failed

Image: Comparing relative focus position measurements  $t_{focus} - z/c$  from a Lasy simulation to the expected evolution.





## 4 What went wrong?

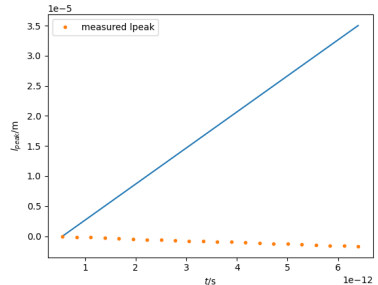
# What went wrong?

## The same simulation in PIConGPU

- Simulation in Easy to the focus region, from there in PIConGPU
  - Result: The flying focus effect cannot be seen here either
- Test also failed

Image: Comparing relative focus position measurements

$z_{focus} - t \cdot c$  from a PIConGPU simulation to the expected evolution.





# What went wrong?

## Checking the shapes of the optical elements

- Multiple differential equations for the shape of the RGD  $\tau_D(r)$ 
  - All result in very similar shapes
- Multiple equations for the shape of the axiparabola  $s_f(r)$ 
  - All follow from the same differential equation (and solve it)
  - All are very similar shapes

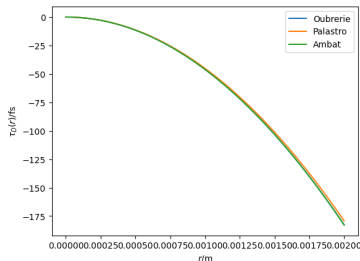


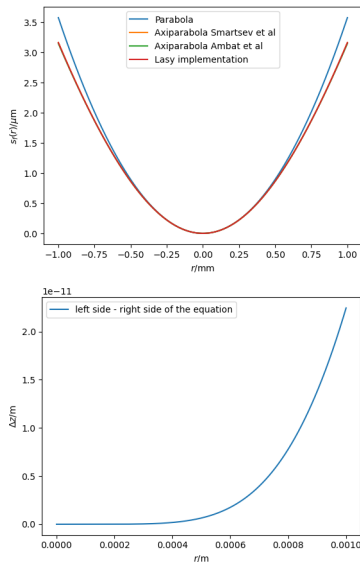
Image: The delay the RGD should achieve according to different sources. The shape of the RGD is similar to this, just stepped in  $\lambda_0/2$  steps. Relative differences  $\approx 1/50$ .

# What went wrong?

## Checking the shapes of the optical elements

- Multiple differential equations for the shape of the RGD  $\tau_D(r)$ 
  - All result in very similar shapes
- Multiple equations for the shape of the axiparabola  $s_f(r)$ 
  - All follow from the same differential equation (and solve it)
  - All are very similar shapes

Images: Top: The shape of the axiparabola according to different sources. Relative differences  $\approx 10^{-2}$ . Bottom: The difference between the two sides of the differential equation the axiparabola is derived from.



# What went wrong?

## Testing all the used modules

- Some successful tests were already shown:
  - A test of Lasy itself
  - A test of the methods to import Lasy pulses into PIconGPU
  - Tests of the RGD

→ These modules work as intended

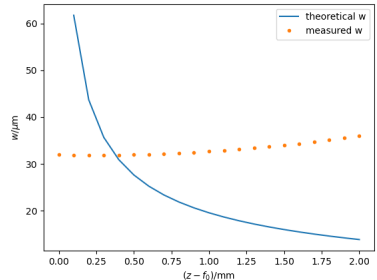
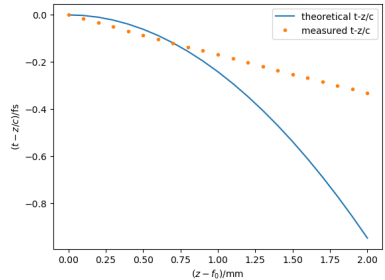
- The field plots corroborate the measured values

# What went wrong?

## Testing the axiparabola alone

- With only the axiparabola differences appear already
  - Here must be the problem
- Using the axiparabola described by Ambat et al [6] gives similar results
- Contacted the Lasy developers to no avail (so far)

Images: Comparing measurements from a Lasy simulation with theoretical values using Ambat et al [6]. Top: Arrival time  $t - z/c$ . Bottom: Beam waist  $w$ .



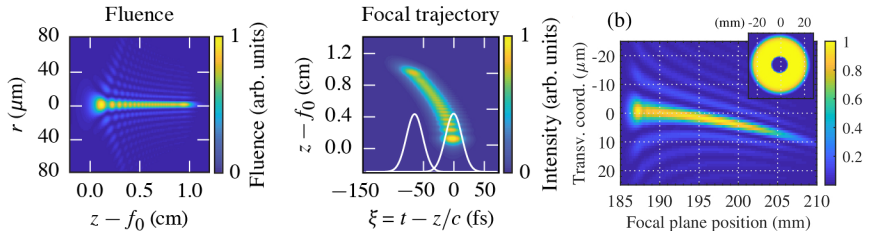


## 5 Summary and Outlook

# Summary

These tests were done:

- All the code was tested with Gaussian laser pulses
  - The shapes of the optical elements were compared to different sources
  - The flying focus pulse was simulated and compared in Lasy and PIconGPU
  - Simulations with just the axiparabola already show differences between theory and measurement
- The Lasy developers were contacted



Simulation results from other papers using only the axiparabola, showing it worked for them without Lasy.

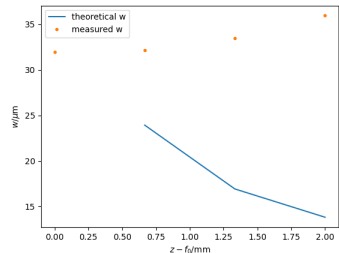
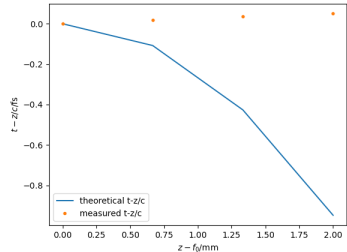
Left (Fluence and Focal trajectory): Ambat et al [6]. Right (b): Smartsev et al [5]

# Further tests after writing

## Using different propagation methods

- Propagation methods tried:
  - Lasy's own angular spectrum propagator
    - Can only handle small numbers of points, useless for us
  - Fresnel chirp-z transform propagator
    - Slow, but can handle larger numbers of points
    - Very similar results to previous tests
- So far, no test successfully showed the flying focus effect

Images: Comparing measurements from a Lasy simulation, using Fresnel chirp-z transform propagation, with theoretical values using Ambat et al [6]. Top: Arrival time  $t - z/c$ . Bottom: Beam waist  $w$ .



- Lazy lasers available in PIconGPU
- LWFA with new laser setups possible
- Specific problems with the axiparabola still need resolving
- Some tests conducted after writing the bachelors thesis, did not help:
  - Trying different propagation methods
  - Initialising the pulse with other pulse profiles (Gaussian, Super-Gaussian, Rect)
  - Using Cartesian coordinates on nodes with 1.5 TB memory

(\*) There's more to explore here.



# References (I)



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Dephasingless laser wakefield acceleration.  
*Phys. Rev. Letters*, 124, 2020.

# References (II)



Slava Smartsev, Clement Caizergues, Kosta Oubrerie, Julien Gautier, Jean-Philippe Goddet, Amar Tafzi, Kim Ta Phuoc, Victor Malka, and Cedric Thaury.

Axiparabola: a long-focal-depth, high-resolution mirror for broadband high-intensity lasers.

*Optics Letters*, 44, 2019.



M. V. Ambat, J. L. Shaw, J. J. Pigeon, K. G. Miller, T. T. Simpson, D. H. Froula, and J. P. Palastro.

Programmable-trajectory ultrafast flying focus pulses.

*Optics Express*, 31(19), 2023.



Lasy 0.6.2 documentation.

<https://lasydoc.readthedocs.io/en/latest>.

Accessed october 2025.



Fabia Dietrich.

Implementation and validation of a method for using experimentally measured laser profiles in laser-plasma simulations with picongpu.

Master's thesis, HZDR, TU Dresden, 2024.

# References (III)



Klaus Steiniger.

Issue 5269 todo in laser refactoring.

<https://github.com/ComputationalRadiationPhysics/picongpu/issues/5269>.

Accessed october 2025.



# Remaining Possible reasons for failure

- The Axiparabola
  - The shape could still be wrong
- The Propagation
  - Possibly the Lasy propagation gives incorrect/incomplete results
- The Findings in the other papers
  - Maybe the Axiparabola does not work at all
  - Maybe it works in a different manner
- The chosen parameters might not fulfill some important approximation condition